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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 08/03/2007 |
| **PI Name:** | Staehelin, Andrew L Ph.D. | | |
| **Project Title:** | Mechanotransduction In Plant Gravity-Sensing Probed By Laser Tweezers And Electron Microscopy | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | UNIVERSITY | **Phone:** | 303-492-8843 |
| **Organization Name:** | University of Colorado | | |
| **PI Address 1:** | MCD Biology | | |
| **PI Address 2:** | Porter Biosciences A3B40 | | |
| **PI Web Page:** | | | |
| **City:** | Boulder | **State:** | CO |
| **Zip Code:** | 80309-0347 | **Congressional District:** | 2 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2001 Space Biology<br>01-OBPR-06 |
| **Start Date:** | 12/01/2002 | **End Date:** | 11/30/2006 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Leitz, Guenther Ph.D. ( University of Colorado ) | | |
| **Grant/Contract No.:** | NAG2-1607 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | The tasks of this project are: 1) to develop a computer controlled laser tweezers microscope in vertical orientation for the non-invasive micromanipulation and functional analysis of gravity-sensing plant cells; 2) to analyze the mechanical properties of the cytoplasm of wild type and mutant gravity-sensing plant cells by optical displacement assays and by video microscopic analysis of statolith sedimentation kinetics; 3) to analyze the initial signal transduction events in plant gravity sensing by combining the optical displacement of statoliths with the simultaneous recording of time-dependent changes of potential second messengers; and 4) to develop precise 3-D models of the cytoskeletal network, the cortical ER network and the nodal ER domains in columella cells of wild type, mutant and transgenic plants. | | |

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| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Many aspects of the cellular and molecular mechanisms of gravity-sensing in higher plants are still unknown. The sites of gravity perception in plant roots are columella cells. The transition of the external gravitational force into the cell body is mediated by high density particles (amyloplasts), which are suspended in the cytoplasm and function as statoliths. The statoliths are involved in the transformation of the physical signal into a biochemical one. The columella cells can transform the mechanical forces acting through statolith sedimentation by a yet unknown mechanism and transmit the sensory information to the root elongation zone. We have identified key aspects of statolith-mediated force transduction in plant gravity sensing. |
| **Task Progress:** | Task progress:<br>Task objective 1: the optical design of the laser tweezers system has been described in detail in the annual reports 2004, 2005 and 2006.<br><br>Task objectives 2, 3 and 4: a detailed description of the results pertaining to these tasks together with many figures was provided in the annual report 2006. The primary objective of work carried out since then has been to integrate the different findings into a major publication, which is now in revision for Plant Cell. The major findings are as follows:<br><br>The starch statolith hypothesis of gravity sensing in plants postulates that the sedimentation of starch-containing amyloplasts (statoliths) in specialized statocytes (columella cells) provides the means for converting the gravitational potential energy into a biochemical signal. How the movement of statoliths is transformed into a biochemical signal is unknown. We have investigated this question by analyzing the sedimentation kinetics of statoliths in the central S2 columella cells of Arabidopsis and by testing the elastic properties of the cortical endoplasmic reticulum (ER) network by laser tweezers. In addition, we have correlated these live cell data with an electron tomography analysis of high-pressure frozen columella cells of Arabidopsis, Nicotiana tabacum and Medicago sativa to characterize the nature of the interactions of the statoliths with the cortical ER. The major findings are: (1) statoliths sediment as aggregates in response to root reorientation up to 135º; (2) within an aggregate, the statoliths approach each other to < 30 nm and, in the contact regions, the statoliths can be deformed to maximize the contact area; (3) the aggregated statoliths exhibit intra-aggregate sliding motions of individual statoliths; (4) root reorientation induces a bouncing motion of the statolith aggregates away from the cortical ER towards the central cytoplasm within 1 s of reorientation; (5) statoliths moved by laser tweezers against the ER boundary experience an elastic lift force upon release from the optical trap; (6) during sedimentation the statolith aggregates tend to move at a distance to the ER boundary and interact only transiently with the membranes of the ER; (7) the weight of statoliths is sufficient to locally deform the ER membranes. We suggest that, in root columella cells, the transduction of the kinetic energy of sedimenting statolith aggregates into a biochemical signal involves a combination of statolith-driven motion of the cytosol and statolith-induced deformation of the ER membranes to activate Ca2+-channels. |
| **Bibliography Type:** | Description: (Last Updated: 08/06/2007) |
| **Abstracts for Journals and Proceedings** | Leitz G, Kang B-H, Schoenwaelder MEA, Staehelin LA. "Statolith sedimentation kinetics and force transduction to the cortical endoplasmic reticulum in gravity-sensing Arabidopsis columella cells." Invited talk, Gordon Conference on Mechanosensory Transduction, Biddeford, ME, July 22-27, 2007.<br>Gordon Research Conference, July 2007. , Jul-2007 |
| **Articles in Peer-reviewed Journals** | Leitz G, Kang B-H, Schoenwaelder MEA, Staehelin LA. "Statolith sedimentation kinetics and force transduction to the cortical endoplasmic reticulum in gravity-sensing Arabidopsis columella cells." Plant Cell, in revision, August 2007. Expected publication December 2007. , Aug-2007 |